



DIGITAL HUMANITIES & RESEARCH SOFTWARE ENGINEERING SUMMER SCHOOL 2024

02-05 JULY 2024

EDINBURGH FUTURES INSTITUTE BUILDING
UNIVERSITY OF EDINBURGH

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Welcome from our Team

The 2024 Digital Humanities and Research Software Engineering Summer School has been co-organised by the Edinburgh Centre for Data, Culture & Society, Cambridge Digital Humanities, King's Digital Lab and the Alan Turing Institute.

The **Centre for Data, Culture & Society** is a specialist hub for data-led research methods within the Edinburgh Futures Institute. It supports a diverse community of practice across the College of Arts, Humanities and Social Sciences at the University of Edinburgh, encouraging experimentation, innovation and skills development. We give tailored support to research groups and projects, offer extensive training opportunities across a variety of applied digital research methods and work to showcase the potential of data-rich and applied digital research.

Cambridge Digital Humanities (CDH) is an interdisciplinary research centre at the University of Cambridge, established to explore the cultural, social and epistemic impacts of digital technologies across various fields in the arts and humanities. CDH collaborates with researchers, cultural institutions, and international partners to develop and deliver research projects with a particular focus on emerging technologies and digital methods. It also offers academic programmes, fostering advancement in areas such as AI, digital libraries and computational humanities.

King's Digital Lab (KDL) at King's College London is a Research Software Engineering team dedicated to enabling high-quality digital research within the Faculty of Arts and Humanities. KDL provides consultancy, development, and training for staff and students. The Lab collaborates widely with higher education institutions, cultural heritage bodies and the creative industries, actively contributing to the digital humanities and research software engineering communities.

The Alan Turing Institute is the UK's national institute for data science and artificial intelligence (AI). It aims to advance world-class research in data science and AI, apply these innovations to address global challenges, and foster skills development and public engagement in these fields. The Institute collaborates with numerous UK universities and partners to drive societal and economic impact through ethical and cutting-edge research and applications.

We would also like to extend our thanks to our speakers who have agreed to showcase good practices in their own work, to the full organising team, who helped create this event.

We hope you will enjoy your time and make the most of your week with us!

Lucia Michielin

CDCS Training Manager

Federico Nanni

Senior Research Data Scientist
at The Alan Turing Institute

Mary Chester-Kadwell

Lead RSE at Cambridge
Digital Humanities

Jonathan Blaney

Digital Humanities Research
Software Engineer at CDH

Neil Jakeman

Senior Research Software
Analyst, King's Digital Lab

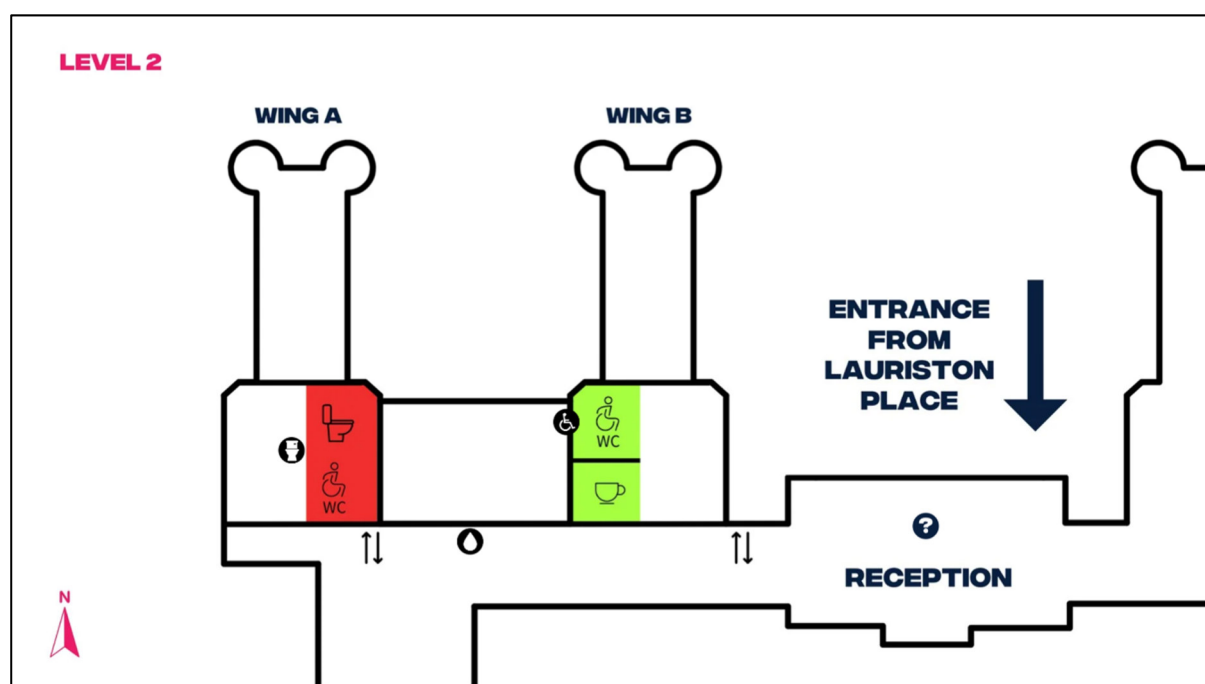
Key Information

The Summer School will take place across the newly refurbished Edinburgh Futures Institute. Registration and teaching will take place on the first floor in **Rooms 1.50 in Wing B**. Our Thursday Reception will take place in the small events space in **Room 4.55 in Wing A**. A Quiet Room has been assigned to **Room 1.63 in Wing A**, in case you need it. Please use this for silent breaks, not for socialising or making calls.

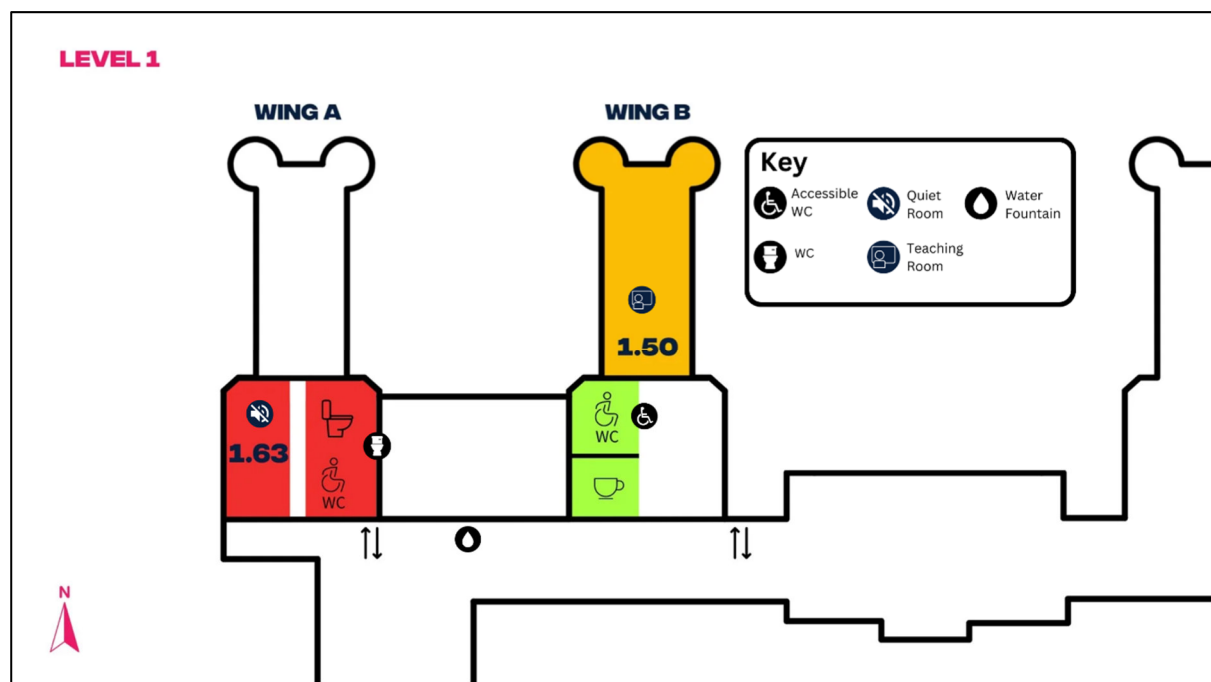
You will enter the building on the 2nd floor from Lauriston Place. Once you enter, Reception will be to your right and the classroom will be on the floor below. During the event, there will be signs around the building to help you navigate the building.

Maps

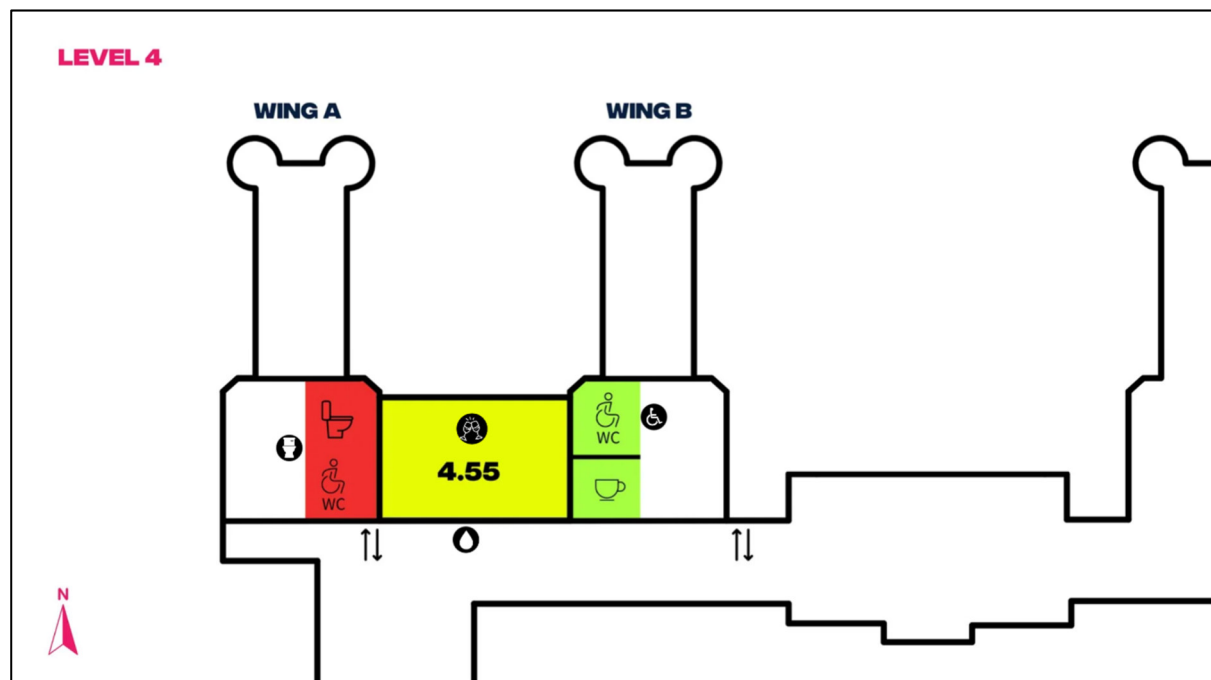
Entrance on Level 2



Teaching Room on Level 1



Thursday Reception on Level 4



Toilet Facilities

On each floor, the main toilet block is in Wing A. All bathrooms are gender-neutral with full privacy. There is also an accessible gender-neutral toilet on each floor in Wing B, past the coffee and tea space.

Water Fountains

Water points to fill up personal water bottles are located in the main corridor of each level.

Getting There

Edinburgh Futures Institute is located next to the Central Campus of the University of Edinburgh and can be accessed from Lauriston Place. The area is well connected to a reliable public transport network.

You can either get there:

- **By bus.** The closest bus stop is at Lauriston Place directly in front of the building. Single bus tickets are £2.00 and can be purchased directly from the driver using contactless payments or paying the exact cash fare.
- **By tram.** This runs from the airport to the city centre. The closest stop to the University is York Place and walking to the campus from there takes around 20 minutes. Single tickets from the airport are £7.50 while single tickets in the city are £2.00 and you have to buy them beforehand, via the trams [app](#), this [website](#), or the ticket machines on the platforms.

A route planner is available on [this website](#).

Registration

On Tuesday, registration will open at 08:40 AM, in **Room 1.50 of the Edinburgh Futures Institute**. Please arrive between **08:40** and **09:00**.

Refreshments and Meals

Lunches will be provided, as well as **morning and afternoon coffee breaks**.

If you need an extra dose of caffeine, or some treats, our recommendations are: Uplands Roast, [Considerit Chocolate](#), [Söderberg The Meadows](#), [Black Medicine Coffee Co](#), [Rudi](#), Press Coffee.

If you are looking for dining options, our recommended spots are: [Ting Thai Caravan](#), [El Cartel](#), [Mother India's Café](#), [City Restaurant](#), [Mosque Kitchen](#), [Pizza Posto](#), [10 to 10 in Delhi](#), [Maki & Ramen](#), [Café Andaluz](#), [Saj Shawerma Grill](#).

Our favourite pubs around the campus are: Doctor's Pub, [The Pear Tree](#), The Dagda Bar, The Dog House, Bennet's Bar, The Devil's Advocate, The Ensign Ewart, [Cloisters Bar](#), [Captain's Bar](#), [Paradise Palms](#).

Format

Each day of the Summer School is structured around four main teaching blocks. These blocks will be a mix between hands-on activities and talks from Research Software Engineers who will present their recent projects.

Each day will be run by a different organising partner. The first day of the Summer School will be organised and presented by University of Edinburgh's Centre for Data, Culture & Society. Workshops on Wednesday will be presented by Cambridge Digital Humanities. Our events for Thursday are led by the team from King's Digital Lab and Friday's talks are organised by the Alan Turing Institute.

Lunch will be served from 12:30 to 13:30 every day. You are welcome to leave the building during this time.

Teams Group and Social Networks

We have created a Teams Group which you can access through this [link](#). There are several channels within the group. You are probably already familiar with the Introduction Channel in which we have asked you to write a few lines about yourself in your acceptance email. In the Social Channel you can message your fellow attendees to make plans after classes and we will use this channel to communicate about our social events throughout the week. The Summer School group is the quickest way to contact attendees and staff and will remain as a virtual shared space for you to keep in touch even after the course.

If you're on Twitter or Instagram, we encourage you to comment live on the Summer School using the hashtags **#DHRSESummerSchool**.

Access to the Wi-Fi

If you are not a staff or student at the University of Edinburgh, but your home institution uses EduRoam, you will be able to access the EduRoam network at the University using the same username and password that you do at your home institution. If you are not from the University of Edinburgh and do not have access to EduRoam, you will be given a temporary passcode on the first day of the Summer School to access our Visit-Ed network. More information about connecting to the Wi-Fi as a guest can be found at this [link](#).

Social Programme

Tuesday 2 July: Ceilidh Club

Join us for a night of incredible folk music and dancing at the Ceilidh Club hosted by Summerhall. No prior experience of ceilidh dancing is necessary, and a caller will walk you through the moves at the start of each dance.

Additional Information:

Time: 20:00 - 22:00

Venue: Summerhall

Tickets are currently sold out for this event but we encourage those of you who were able to buy tickets in advance to attend together!

Thursday 4 July: Reception

After the conclusion of Thursday's activities, we will have a drinks reception. This is a great chance to meet and socialise with fellow Summer School attendees, alongside your lecturers and instructors in an informal setting.

Additional Information:

Time: 17:30 – 19:00

Location: Small Events Space Room 4.55

Attendance is free and complimentary food and drinks will be available.

Friday 5 July: Pear Tree Garden Drinks

To celebrate the end of a very busy week, if the weather permits, we'll be heading to the Pear Tree Beer Garden.

Additional Information:

Time: 17:30 - 19:00

Venue: Pear Tree Pub

Food and drink are available at the attendee's own expense.

Our Instructors and Presenters

Centre for Data Culture and Society

Aislinn Keogh

Aislinn is a PhD student in the Centre for Language Evolution at the University of Edinburgh, supervised by Professor Jennifer Culbertson and Professor Simon Kirby. She uses behavioural experiments and computational modelling to study the evolution of linguistic structure, focusing on cognitive demands in real-time language processing and production. Her research, funded by the Scottish Graduate School of Social Science, investigates pressures and biases in language production, trade-offs between production and comprehension biases, and variations in mental representation of linguistic structure among individuals.

Alexis Pister

Alexis Pister is a data visualization specialist with a Ph.D. from Paris-Saclay University, focusing on network visualization in historical records. He also holds a Master's degree in Bioinformatics and modeling from INSA Lyon. At present, Alexis leverages his expertise to develop interactive applications and data visualization interfaces for data-intensive academic research projects spanning arts, humanities, and computer science, facilitating in-depth data exploration and effective communication of research findings.

Ed MacKenzie

Ed MacKenzie brings two decades of software development experience across commercial and public sectors, complemented by a deep passion for history, economics, and humanities. His expertise in digital humanities spans roles at prestigious institutions like Kings College London, University of Sheffield, University of Cambridge, and University of Oxford. At the Edinburgh Futures Institute and CDCS, Ed's focus is on advancing digital methodologies for research, overseeing projects like 'tei-service' for text transcription support and modernizing historical projects such as the 'Medieval Dedications of Scottish Saints' and the 'Survey of Scottish Witchcraft'.

Evan Morgan

Evan Morgan is a Research Software Engineer at the University of Edinburgh's Design Informatics department, specializing in applying data-driven technologies to enhance innovation across various research initiatives. With extensive experience spanning industry and academia, he completed a PhD at Queen Mary University of London focusing on sensor technologies to enhance non-verbal interactions among musicians. Since joining Edinburgh in 2018, Evan has contributed significantly to projects like ENHANCE and IDEAL, focusing on designing interventions for behaviour-based energy reduction within the School of Informatics.

Lisa Otty

Lisa Otty leads the Centre for Data, Culture & Society at the University of Edinburgh, overseeing its strategic direction and diverse research support services. With extensive experience in software development, research services, and digital humanities, she supports innovation and co-creative work across the university. Previously a lecturer in Digital Humanities, Lisa is dedicated to enabling researchers in the arts and humanities to engage critically and creatively with digital methods.

Lucia Michielin

Lucia's manages and develops the Centre for Data, Culture & Society Training Programme, a programme which focuses on identifying and providing a variety of training on applied digital research to respond to researchers' needs across the College of Arts, Humanities and Social Science. She has considerable experience in designing and delivering training courses focused on data skills and digital research methods, and can apply specialist technical expertise and knowledge to developing a digital research methods training programme.

Sarah Schöttler

Sarah is a PhD candidate in the Visual+Interactive Data group at the University of Edinburgh, researching responsive geographic visualization – how to make maps, cartograms and other geographic visualizations work across devices and screen sizes. Prior to this, she completed an MSc in Design Informations at the University of Edinburgh and a BSc in Technology and Liberal Arts & Sciences at the University of Twente in the Netherlands. She also teaches workshops and short courses on programming, visualization and other digital skills as a CDCS Training Fellow.

Will Lamb

Will Lamb, originally from Baltimore, is a Professor in Gaelic Ethnology and Linguistics at the University of Edinburgh, known for his extensive work in music, linguistics, traditional narrative, and language technology. He completed his PhD on Scottish Gaelic register variation, which was published in 2008, and has significantly contributed to Gaelic NLP and the successful music programme at Lews Castle College. His research includes notable work on formulaic language, Gaelic grammatical description, and Natural Language Processing, and he is currently pursuing an MSc in Speech and Language Processing.

Cambridge Digital Humanities

Estara Arrant

Estara Arrant is a multidisciplinary scholar specializing in Semitic languages, machine learning, and medieval Jewish and Muslim textual traditions. Currently a research associate at Trinity Hall and involved in the ERC-funded TEXTEVOLVE project, she adapts bioinformatics algorithms to study the evolution of medieval Targum manuscripts. Her extensive academic background includes a PhD from the University of Cambridge focusing on codicology and linguistics of Torah manuscripts from the Cairo Genizah, employing machine learning to analyse linguistic features influenced by Arabic and Aramaic.

Johnathan Blaney

Jonathan is Digital Humanities Research Software Engineer at CDH. Previously he worked on Digital Humanities projects at the Institute of Historical Research, University of London, and before that at the Oxford Digital Library in the Bodleian. Going back even further, he worked as a lexicographer at Oxford University Press, across the range of English dictionaries and thesauruses.

Lidea Shahidi

Lidea Shahidi is a postdoctoral research associate at the MRC Cognition and Brain Sciences Unit, with a PhD from Duke University, where she developed speech enhancement strategies for cochlear implant recipients. Her current research focuses on novel speech coding strategies to improve clinical outcomes for cochlear implant users, employing computational analyses and behavioural experiments. Additionally, she has interests in auditory scene analysis, speech enhancement, and speech processing, and she has taught workshops on programming, psychoacoustics, and signal processing.

Mary Chester-Kadwell

Mary is a Senior Software Engineer at Cambridge University Library. Originally from a humanities background, she has a PhD in the landscape archaeology and material culture of early medieval England using computational methods. For CDH Labs, Mary advises researchers on developing the technical aspects of their projects and works on a wide variety of software. She teaches coding and software engineering practices for CDH Learning and sits on the teaching team for the MPhil in Digital Humanities. Mary is an active member of the Research Software Engineering community.

Michael Hawkins

Dr Michael Hawkins has been working in Digital Humanities since completing his PhD on Thomas Willis's neurocartography of the passions at Imperial College London in 2004. He has served as

Technical Director of numerous digital humanities projects. His main areas of interest are editing and encoding materials for historical research. His specialities include TEI, XSLT, xQuery, XPath, Schematron, Python, JavaScript and CSS.

King's Digital Lab

Elliot Hall

Elliott Hall is a Senior Research Software Engineer, Creative Technologist, and Writer. He creates immersive family shows using AR, embedded systems, and live performance, and is known for projects like the Digital Ghost Hunt and the AI installation Room is Sad. As a full stack developer, he collaborates with academic partners using technologies such as Django, Python, SQL, and Javascript, and he is also the author of The Strange Trilogy.

Neil Jakeman

Neil, a Senior Research Software Analyst, joined the Department of Digital Humanities in 2011, initially focusing on spatial analysis and broadening his interests to various departmental topics. He has led technical development on several projects, including creating a database on the origins of the Celtic language and a tool for studying medieval French manuscripts. Currently, he cultivates new projects and relationships, particularly in digital creative practices involving emerging technologies like AR, VR, and real-time engines, while also being an accomplished 3D modeller and photographer.

Oscar Seip

Oscar Seip is based in Manchester and joined the Software Sustainability Institute in December 2023 as a Research Software Community Officer. Originally from the Netherlands, he previously worked within the fields of arts and humanities as an academic researcher and theatre producer. Additionally, Oscar holds the status of Associate Fellowship from Advance Higher Education and acquired several qualifications in software development.

The Alan Turing Institute

Ed Chalstrey

Ed is a Research Data Scientist in The Alan Turing Institute's Research Engineering Group (REG), with a background in Bioinformatics. In the Research Engineering Group, he works on various academic

projects providing support for collaborators via his skills in research software engineering and data science.

Federico Nanni

Federico is a Senior Research Data Scientist at The Alan Turing Institute, leading the Fundamental AI group within the Research Engineering Group. He specializes in Natural Language Processing and Information Retrieval, particularly in Digital Humanities and Computational Social Science. Federico holds a PhD in History of Technology from the University of Bologna, where he focused on web archives in historical research. Before joining the institute, he worked for five years with the Data and Web Science Group at the University of Mannheim. Additionally, he has been a visiting researcher at the Foundation Bruno Kessler, Aarhus University, and the University of New Hampshire.

Kaspar Beelen

Kaspar Beelen is a digital historian with a PhD in History and Linguistics from the University of Antwerp who applies machine learning to humanities research. He has conducted postdoctoral research on parliamentary discourse and held academic positions at the University of Amsterdam and the Alan Turing Institute, focusing on digitized heritage and media studies. His work spans diverse topics, including gender representation in parliament, affect in political language, bias in digitized newspapers, large-scale map analysis, machine animacy history, and place name detection in text.

Rosie Wood

Rosie Wood is a Junior Research Data Scientist at the Alan Turing Institute who joined the Living with Machines project in 2023. Her background is in Computational Chemistry and she has been working as one of the core developers on the MapReader package for a year and a half.

Stavros Angelis

Stavros Angelis is a senior technical officer/digital officer at the Maynooth University Arts and Humanities Institute. He studied Archives and Library Science in the Ionian University, Greece, where he also received his MSc in Information Science. Stavros has worked extensively on defining, implementing, extending, and transforming metadata schemas and creating mappings and crosswalks across various information structures. He has always been a part of interdisciplinary teams, working independently and/or collaborating with people of different backgrounds.

Programme

Timetable

Tuesday		Wednesday		Thursday		Friday	
Centre for Data, Culture & Society		Cambridge Digital Humanities		King's Digital Lab		The Alan Turing Institute	
08:40-09:10	Registration	09:00-09:15	Introduction and Society of RSE presentation	09:00-09:30	Introduction & SSI talk	09:00-09:30	Introduction & DHRSE IEUK network talk
09:10-09:30	Welcome and Introduction	09:15-10:30	DH & RSE at Cambridge	09:30-10:30	Digital creative approaches for research and engagement	09:30-10:30	MapReader, open-source software library for analysing large map collections
09:30-10:30	Project Deep Dive						
10:30-11:00	Coffee Break	10:30-11:00	Coffee Break	10:30-11:00	Coffee Break	10:30-11:00	Coffee Break
11:00-12:30	DH&RSE at Edinburgh	11:00-12:30	Sustainable code for research	11:00-12:30	Digital creative approaches for research and engagement	11:00-11:45	DH & RSE practices: the Seshat Project
						11:45-12:30	DH & RSE practices: Living with Machines
12:30-13:30	Lunch	12:30-13:30	Lunch	12:30-13:30	Lunch	12:30-13:30	Lunch
13:30-15:00	Data visualisation with R	13:30-15:00	Applying Best Practices for Sustainable Code	13:30-15:00	Constructing an Immersive experience from a research scenario	13:30-15:00	Collaborative coding
15:00-15:30	Coffee Break	15:00-15:30	Coffee Break	15:00-15:30	Coffee Break	15:00-15:30	Coffee Break
15:30-17:00	Custom data visualisation with D3	15:30-17:00	Applying Best Practices for Sustainable Code	15:30-17:00	Constructing an Immersive experience from a research scenario	15:30-17:00	Collaborative coding
20:00-22:00	Ceilidh Club			17:30-19:00	Reception	17:30-19:00	Drinks at the Pear Tree

Day 1 Tuesday – Centre for Data, Culture & Society

Welcome and Introduction

Dr Lucia Michielin and Dr Lisa Otty will kick off the summer school and will introduce the Centre for Data, Culture & Society day with an overview of the Edinburgh University DH& RSE landscape.

Project Deep Dive

Coding my way out of a box: Achieving emancipation and collaboration through the Digital Humanities

Will Lamb - Professor in Gaelic Ethnology and Linguistics, University of Edinburgh

Prof Will Lamb explores the intersections between Celtic Studies and Digital Humanities at the University of Edinburgh, highlighting his transition from ethnology and linguistics to Natural Language Processing (NLP). This talk delves into the methodologies and research initiatives at the University of Edinburgh that integrate computational techniques with humanities scholarship. Attendees will gain a deeper understanding of how Research Software Engineering (RSE) practices are enhancing the study of cultural heritage and linguistics here, providing foundations for future innovations. This session offers inspiration and practical insights for those looking to deepen their engagement with Digital Humanities and RSE.

DH & RSE at Edinburgh

Ed MacKenzie - Research Technologist, Edinburgh Future Institute

Alexis Pister - Data Visualization Engineer, Edinburgh Future Institute

Evan Morgan - Research Software Engineer, Design Informatics

A panel with three engineers within the University of Edinburgh that will explore what it means to work as a Research Software Engineer within the Humanities and Social Sciences College. Ed Mackenzie has twenty years of experience in software development in both the commercial and public sectors. Alexis Pister is a data visualisation research engineer working at the Edinburgh Future Institute and the Vishub. Evan Morgan is a Research Software Engineer in Design Informatics at the University of Edinburgh.

Data Visualisation with R

Lucia Michielin - Digital Skills Training Manager, Centre for Data, Culture & Society

Aislinn Keough - Training Fellow, Centre for Data, Culture & Society

This workshop will guide participants from basic to advanced data visualization techniques using RStudio and the powerful ggplot2 package, grounded in the principles of the Grammar of Graphics. Attendees are going to explore creative ways of plotting sentiment analysis results and mapping whisky consumption in Scotland.

Custom Data Visualisation with D3

Sarah Schöttler - Training Fellow, Centre for Data, Culture & Society

Aislinn Keogh - Training Fellow, Centre for Data, Culture & Society

In this workshop, you'll learn how to create custom data visualizations using D3.js when standard visualizations fall short. We'll start with a sketching exercise on visual variables using a data set, followed by a hands-on implementation in Observable, covering how the platform works and applying our sketches.

Day 2 Wednesday – Cambridge Digital Humanities

Welcome and Society of Research Software Engineer Talk

Mary Chester-Kadwell will be introducing us to Day 2 of the Summer School. Mary is a lead Research Software Engineer at Cambridge Digital Humanities. She will also introduce the work of the Society of Research Software Engineering that sponsors the Summer School. The Society aims to support the recognition of the vital role that software plays in research and support those who develop research software in all its forms.

DH & RSE in Cambridge:

Encouraging Sustainability in Digital Humanities Projects

Michael Hawkins - Senior Developer, Cambridge University Library

Sustainability can be challenging for digital humanities projects. Fixed-term funding means that that you're often locked within an endless cycle of fund-seeking. Unless you build and manage your own infrastructure, your existence also relies on the good-graces and continued support of your local support team, who are likely understaffed and overstretched. Despite these challenges, there are strategies that you can adopt to encourage the long-term sustainability of your project. Trained as a historian of early modern science and medicine, Michael Hawkins has been the Technical Director of many high-profile digitisation projects over the past twenty years, including The Newton Project, The Casebooks Project and the Darwin Correspondence Project.

Bridging the Gap between Philology of Middle Eastern Texts and Data Science

Estara Arrant – Postdoctoral Research Associate, Cambridge University Library

The linguistic and historical study of the Medieval Middle East is advanced by using data science tools and methods. In this talk I will describe how I have embraced the possibilities, challenges and experiences of working in digital humanities as a data scientist and research software engineer. For my doctoral thesis, I mined physical and linguistic data from a large Hebrew manuscript collection to find patterns in material and linguistic culture from the 10th-15th centuries. In my current position, I develop and modify tools from biological sciences to study the evolution of a text as it is copied over time and place. In the future, I plan to explore the interplay between language and society using a combination of natural language processing and categorical data analysis/clustering for a medieval corpus of Judaeo-Arabic letters. Through these projects, I have learnt practices and approaches for sustaining the methods, code and data I have developed.

Careers Panel

This careers panel will be a conversation about research software engineering between Dr. Mary Chester-Kadwell, Jonathan Blaney, Dr. Estara Arrant, and Dr. Michael Hawkins. Dr. Mary Chester-Kadwell is a Senior Software Engineer at Cambridge Digital Humanities. Jonathan Blaney is a Digital Humanities Research Software Engineer at Cambridge with previous experience in projects at the Institute of Historical Research at the University of London and at the Oxford Digital Library in the

Bodleian. Dr. Estara Arrant is a Postdoctoral Research Associate at the Cambridge University Library and Dr. Michael Hawkings is a Senior Developer at Cambridge University Library.

Sustainable Code for Research:

Introduction to Best Practices for Sustainable Code

Dr Mary Chester-Kadwell – Lead RSE, Cambridge Digital Humanities

There is no single, definitive thing that makes code ‘good’ or ‘sustainable’. Rather, there are categories of ‘best practices’ you can choose for your project, depending on what you are trying to achieve and how much time you have available. The ‘one simple trick’ is to be selective and make an informed, deliberate choice. This talk covers a range of options for improving the sustainability of your research code, such as: ensuring that code runs reliably now and in the future; showing how (and why) the code works for yourself and others; and facilitating access to code when sharing with others. Topics include version control (the foundation of everything else), dependency management, documentation of various types, refactoring, testing, open source, publishing and where to look for further resources.

Evaluating GitHub Repositories

Jonathan Blaney – Digital Humanities RSE, Cambridge Digital Humanities

In this session we will look at some GitHub repositories to see what we can learn about good (and not so good) practices for writing sustainable code. Is the repository well organised and clearly documented? Is the code itself written with an emphasis on readability and consistency? Would you want to use this repository or steer well clear? The discussion aims to prime attendees for the hands-on sessions in the second half of the day. Jonathan supports students on the MPhil in Digital Humanities for Cambridge Digital Humanities. Previously he worked on DH projects at the Institute of Historical Research, University of London, and before that at the Oxford Digital Library in the Bodleian.

Applying Best Practices for Sustainable Code

Lidea Shahidi – RSE Methods Fellow, Cambridge Digital Humanities

This session will give participants hands-on experience with coding design principles and preparing code for sharing. These practices help to ensure coding projects are reliable, reusable and understandable for you and for others. The session will begin with a demonstration of sustainable coding practices and their practical implementation. The demonstration will be followed by group work in which participants will apply the concepts to an example project, and will include tasks concerning formatting, documentation, refactoring, and publishing code.

Dr Lidea Shahidi is a Research Software Engineering Methods Fellow at CDH where she helps researchers in the humanities apply best practices in coding. Concurrently, Lidea is a postdoctoral research associate at the MRC Cognition and Brain Sciences Unit in Cambridge, and conducts research on hearing, speech perception, and cochlear implants.

Day 3 Thursday – King's Digital Lab

Introduction & SSI Talk

Neil Jakeman and Elliot Hall will be introducing Day 3 of the Summer School which will be hosted by King's Digital Lab. Neil Jakeman is a Senior Research Software Analyst and Elliot Hall is a Senior Research Software Engineer both at King's Digital Lab.

Introduction of SSI and its Fellowship Programme

Oscar Seip - Research Software Community Officer for SSI

The Software Sustainability Institute (SSI) supports sustainable software practices through its Fellowship Programme, which aims to improve and promote good computational practice across all research disciplines and support those who are doing this important work. In this talk, you will learn more about the SSI and the opportunities of the Fellowship Programme.

Digital Creative Approaches for Research and Engagement

Neil Jakeman - Senior Research Software Analyst, King's Digital Lab

Elliot Hall - Senior Research Software Engineer, King's Digital Lab

Discover innovative methods to enhance your research and engagement strategies using digital tools in this hands-on workshop. We'll explore cutting-edge techniques that integrate creativity with rigorous research, enabling you to visualize, analyse, and present your data in compelling ways.

Creative Technologist role in impact and engagement

We will discuss the role of digital creativity in creating impact and engagement in research projects and look at how our typical software development life cycles (SDLC) might need to be adapted to deal with a different sort of output. Morning sessions will look at the GLoW3 exhibition as a case study of collaboration with artists.

Creation and Data types in 3D

Understanding data types in 3D technology requires a new vocabulary. We discuss the main data types and the considerations we should make when preparing data for different purposes – e.g. game engine development, or 3D animated render. We will use a capture some 3D data and evaluate it objectively and look at platforms for sharing this data.

Constructing an Immersive Experience from a Research Scenario

Neil Jakeman - Senior Research Software Analyst, King's Digital Lab

Elliot Hall - Senior Research Software Engineer, King's Digital Lab

Transform your research into captivating immersive experiences in this hands-on workshop. Gain practical skills in creating immersive content that enhances understanding and engagement, making your research compelling and accessible to a broader audience.

We suggest a typical scenario and request from a small heritage organisation wanting to share their collections more effectively through digital means. In groups, you will develop a strategy to address their requirements and an implementation plan. If time and resources allow, we may move on to proof-of-concept development.

Day 4 Friday – The Alan Turing Institute

Introduction & DHRSE IEUK Network Talk

Federico Nanni introduces the final day of the Summer School hosted by The Alan Turing Institute. Federico is a Senior Research Data Scientist at The Alan Turing Institute.

Research Software Engineering in the Arts and Humanities: a community-driven approach

Stavros Angelis - Senior Technical Officer/Digital Officer, Arts & Humanities Institute, Maynooth University

Why do we need and how can we secure Research Software Engineering capability in the Arts and Humanities? We present a Community Interest Group that aims to offer a forum for the Research Software Engineering (RSE) community working and collaborating in Arts and Humanities across UK and Ireland for focused discussion, expertise and knowledge exchange, action, advocacy and policy making towards the development of AH RSE capability.

DH & RSE practices

MapReader, open-source software library for analysing large map collections

Rosie Wood – Research Data Scientist, The Alan Turing Institute

MapReader (<https://github.com/Living-with-machines/MapReader>) is an end-to-end computer vision pipeline for image classification, specifically aimed at researchers looking to work with images of historic maps. It was initially developed as part of the Living with Machines (LwM) project in 2018 and so, when I joined in 2023, MapReader was already a well-defined software package. However, it needed significant work to enhance user-friendliness, integrate new features/functionality, and implement better testing and documentation.

Unlike many of the LwM team members, my background is in Computational Chemistry rather than Digital Humanities (DH) and so joining a DH project was a steep learning curve. Likewise, the transition from research to a Research Software Engineer (RSE) role was a significant change and took some getting used to. In both cases, there were many new concepts to learn, new ways of working to adapt to and many, many, acronyms to contend with!

I have now been working as one of the only core developers on the MapReader package for 1.5 years and in this talk will discuss my experiences and lessons learned throughout this time.

DH & RSE practices: the Seshat Project

Ed Chalstrey – Research Data Scientist, The Alan Turing Institute

The “Seshat Global History Databank” tracks over 150 data variables related to technology, language, religion, warfare, economics and more, across roughly 800 historical polities (nations, empires etc), to aid historians conducting data-driven research. As an RSE at The Alan Turing

Institute, I have been working on an AHRC funded project that aims to bring the benefits of RSE to research in the arts and humanities. As part of this, I was tasked with integrating a GeoJSON shape dataset of historical polities covering the last 5,000 years into Seshat, which involved extending the Django website code with an interactive world map. I will demonstrate the functionality of the world map on the Seshat website, offering explanations on the decision processes in data visualisation, such as colour choices and how to handle different types of data variable (e.g. boolean, categorical). I will discuss the technical challenges, including loading over 10,000 geospatial shape data objects in the browser and how they are handled. I will also describe the experience in bringing best practices in software engineering to a mixed team of engineers and data-driven historians on a mature research project running for over a decade, and the challenges of doing so.

DH & RSE Practices: Living with Machines

Kaspar Beelen – Technical Lead, School of Advanced Study

In this talk, we will discuss collaboration in large-scale digital humanities projects, primarily from the perspective of "humanists who can code." How can we efficiently operate in large interdisciplinary projects and work with scholars from radically different backgrounds? Our focus will be on the Living with Machines project, a radical collaboration involving the Alan Turing Institute and the British Library, among others. We will explore the types of collaboration that worked, but also discuss what remained challenging, and why.

Collaborative Coding

Federico Nanni - Senior Research Data Scientist, The Alan Turing Institute

Collaboratively writing and reviewing code is a crucial skill for working as part of a team. Keeping track of outstanding issues and proposed solutions, as well as ensuring compatibility of code written by different team members becomes challenging as the team size grows. In this hands-on workshop, we will review the "Git flow" framework for efficiently organising and tracking collaborative programming, and practice creating GitHub Issues, Git branches, and Pull Requests.